



The effect of CO₂ on a cobalt-based catalyst for low temperature Fischer–Tropsch synthesis

Y. Yao, X. Liu, D. Hildebrandt*, D. Glasser

Centre of Material and Process Synthesis, School of Chemical and Metallurgical Engineering, University of the Witwatersrand, Johannesburg, South Africa

ARTICLE INFO

Article history:

Received 5 December 2011

Received in revised form 2 April 2012

Accepted 16 April 2012

Available online 24 April 2012

Keywords:

Fischer–Tropsch synthesis

Hydrogenation

Carbon dioxide

Carbon monoxide

Fixed bed reactor

Cobalt catalyst

ABSTRACT

A series of Fischer–Tropsch synthesis (FTS) experiments, which entailed repeatedly switching between a CO (CO/H₂/N₂) and a CO₂ (CO₂/H₂/N₂) feed, were conducted in a fixed bed reactor over a cobalt-based catalyst. It is worth noting that the effect of the CO₂ on the properties of a cobalt-based catalyst was very small under the reaction conditions we chose. There was no apparent catalyst deactivation at reaction temperatures of 180 °C and 200 °C when we continually alternated between the CO and CO₂ feeds.

We observed dramatic changes in the catalyst activity and product selectivity for CO₂ hydrogenation before and after the initial FTS for CO feed at 180 °C. In addition, during the initial CO hydrogenation on the cobalt catalyst, both the olefin and paraffin formation rates suddenly changed from one pseudo-stable state to another. These differences may have been caused by liquid products, whether deposited on the catalyst surface or in the catalyst pores during CO FTS.

A mild catalyst deactivation was observed at the operating temperatures of 210 °C and 220 °C, respectively. According to the comparison we made between the conversion of the feed gases and the product formation rates for paraffin and olefin, and our speculations concerning possible side reactions, we conclude that the catalyst deactivation is possibly attributable to the re-oxidation by water.

© 2012 Elsevier B.V. All rights reserved.

1. Introduction

The low-temperature (180–250 °C) Fischer–Tropsch synthesis (FTS) over either iron or cobalt catalysts, producing high molecular mass linear waxes, which in turn can be hydro-cracked to produce diesel of exceptionally high quality [1–3]. Cobalt is considered the most suitable metal for the low-temperature FTS of long chain hydrocarbons because its activity and selectivity to linear paraffins are high, and its water–gas shift (WGS) activity is low [1,4,5].

As the cobalt catalysts used in FTS are relatively expensive (compared to the cost of iron), they need to have a high metal dispersion and long life to be able to offer a good balance between cost and performance [6,7]. This is why catalyst deactivation is a major challenge in cobalt-based FTS [6–8]. The oxidation of cobalt metal to cobalt oxide by the product water, the most abundant byproduct of FTS, has long been believed to be a major cause of the deactivation of supported cobalt FTS catalysts [6,7,9–11]. Owing to the low activity a cobalt catalyst has for WGS, CO₂ is not the major byproduct. Nevertheless, in some cases CO₂ may be a significant component in the syngas obtained from biomass and coal [3,12]. It is therefore necessary to investigate the effect of CO₂ (as an oxidizing agent) on cobalt-based low-temperature FTS.

* Corresponding author. Tel.: +27 (0)11 7177527; fax: +27 (0)11 7177604.

E-mail address: diane.hildebrandt@wits.ac.za (D. Hildebrandt).

Until recently, the effect of CO₂ on cobalt-based FTS has remained controversial. Some researchers [3,13–15] believe that CO₂ behaves as an inert diluent in the syngas feed at temperatures below 220 °C for FTS over cobalt-based catalysts. Zhang et al. [15] claimed that the catalyst deactivates more rapidly for the conversion of CO than for CO₂, even though the H₂O/H₂ ratio is at least two times greater for the conversion of CO₂ in cobalt-based FTS. However, Kim et al. [16] concluded that the presence of CO₂ in the feed gas affects the rate of catalytic hydrogenation of CO as well as the product distribution, and that CO₂ acts as a mild oxidizing agent on reduced Co/γ-Al₂O₃ at 220 °C and 20 bar. Riedel and Schaub [17] also found that CO₂ had a negative effect on both the FT reaction rate and deactivation with a catalyst comprising Co–La–Ru–SiO₂. A cobalt catalyst used with a temperature of 220 °C for FTS may also cause WGS activity and an increase in methanation rates [3]. The technique most commonly applied when studying the effect of CO₂ is the co-feeding of CO₂ in the feed gas during low-temperature FTS [13–19], but relatively little of the published research [13–21] has dealt with the effect of CO₂ on cobalt-based FTS.

Furthermore, the chemical utilization of CO₂ as a carbon resource is important from both the economic and environmental standpoints [22]. There have been various attempts to transform carbon dioxide into hydrocarbons, mainly by using catalysts that have been proven to be active in FTS, such as Ni, Ru and Co [23]. Although the need for CO₂ separation before the syngas is used

in FTS is mentioned in the open literature [17,24], recent process development studies suggest a potential advantage in not removing the CO_2 before synthesis takes place. This can be done if the conditions are CO_2 tolerant or if CO_2 is hydrogenated along with CO in the FT reactor. The omission of the separation step is desirable not only to make the process more economical but also to contribute to sustainable development.

In the research described in this paper, we investigated the effect of CO_2 on cobalt-based low temperature FTS by means of a series of new experiments. The nature of the test was to repeatedly switch between the two feed gases, a CO_2 mixture ($\text{CO}_2/\text{H}_2/\text{N}_2$) and a CO mixture ($\text{CO}/\text{H}_2/\text{N}_2$), which were introduced into a micro fixed bed reactor (FBR) for FTS at 180–220 °C, 20 bar (gauge) and 30 ml(NTP)/(min·gcat) over a Co/ TiO_2 catalyst. This provided a means of ensuring that any changes we observed were due to the synthesis gas itself, and not because of permanent or long term changes to the surface or properties of the catalyst.

2. Experimental

2.1. Catalyst preparation

The Co/ TiO_2 catalyst used in this series of experiments was prepared by impregnating TiO_2 with a cobalt nitrate solution. TiO_2 (Degussa P25) was mixed with distilled water in a mass ratio of 1:1, and dried in air at 120 °C for 1 h. The support was then calcined in air at 400 °C for 16 h [25]. After calcination the support was crushed and sieved, and the particles with diameters between 0.5 and 1 mm were selected for use. The support was then impregnated with a sufficient quantity of cobalt nitrate ($\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$) solution to give a cobalt metal loading of 10% by mass. Thereafter the support was dried in air at 120 °C for 16 h, and then calcined in air at 400 °C for 6 h to allow it to decompose and transform from cobalt nitrate to cobalt oxide.

2.2. Experimental set-up and procedure

We loaded 1 g of catalyst into the FBR, and performed the reduction at atmospheric pressure with H_2 (Afrox (African Oxygen) Ltd., 99.999%) for 24 h. The reduction temperature and the flow rate were 350 °C and 60 ml(NTP)/(min·gcat), respectively.

Once the reduction was completed, we allowed the reactor to cool down to room temperature. The CO_2 syngas (hereafter referred to as the CO_2 feed), composed of $\text{CO}_2:\text{H}_2:\text{N}_2 = 23\%/67\%/10\%$ whose N_2 served as an internal standard for mass balance calculations, was initially introduced into the reactor at a flow rate of 30 ml(NTP)/(min·gcat) (Fig. 1a). The reactor pressure was slowly increased to 20 bar (gauge), after which the temperature was gradually raised to 180 °C. The pressure and temperature were allowed to stabilize, and these operating conditions were maintained in a constant state for 72 h, during which the composition of the tail gas was monitored. Next, the feed gas was switched from CO_2 feed to the CO syngas (designated as the CO feed), which consisted of

$\text{CO}:\text{H}_2:\text{N}_2 = 30\%:60\%:10\%$, with N_2 as an internal standard for mass balance calculations (Fig. 1b). The new reaction conditions retained the same constant pressure, temperature and flow rate as those for the CO_2 feed. These were maintained for 72 h while the tail gas composition was monitored. After that, the feed gas was switched back to CO_2 feed with the same operating conditions in terms of the constant pressure, temperature and flow rate. Each of the feed gases was interchanged around four to five times, under the same operating conditions. The same sequence was repeated at 200 °C, 210 °C and 220 °C.

The monitored results for all the reaction conditions we conducted showed that 72 h was enough time to stabilize the reactor. The simplified flow scheme for the FTS experiments is given in Fig. 1, and the experimental reaction conditions are shown in Table 1.

2.3. Product analysis

The tail gas was analyzed every 1.5 h using an online DANI GC. Two thermal conductivity detectors (TCD) were used to analyze H_2 , N_2 , CO, CO_2 and CH_4 , while a flame ionization detector (FID) did the same for the gas phase hydrocarbons. The wax and liquid products were collected in a hot trap (kept at 150 °C) and cold trap (maintained at room temperature). The oil and wax products were analyzed by an off-line GC at the end of the mass balance for each run.

3. Results

3.1. The effect of CO_2 on the catalyst activity and selectivity for CO and CO_2 hydrogenations

3.1.1. Reactant conversion and product selectivity

The reactant conversions and product selectivity for both the CO and CO_2 feeds during 2700 h on-stream are given in Fig. 2. The data in Fig. 2 show that the CO and CO_2 are readily hydrogenated on a cobalt-based catalyst. For the CO feed, the data presented indicate the following.

- The CO conversion, CH_4 selectivity and C_{2+} selectivity did not change much during the switching between the CO and CO_2 feeds at each operating temperature.
- The conversions increased when the temperature rose from 180 °C to 220 °C. In the mean time, the CH_4 selectivity increased slightly from around 8% to 12%, contrarily, the C_{2+} selectivity declined with increasing reaction temperature.
- On the other hand, the results for the CO_2 feed follow a slightly different pattern:
- Similarly to the CO conversion, the CO_2 conversion improved with an increase in temperature. At the lower temperature of 180 °C, the catalyst reactivity for CO_2 was close to that of CO. However, when we increased the reaction temperature from 200 to 220 °C, CO_2 demonstrated a lower reactivity than CO.

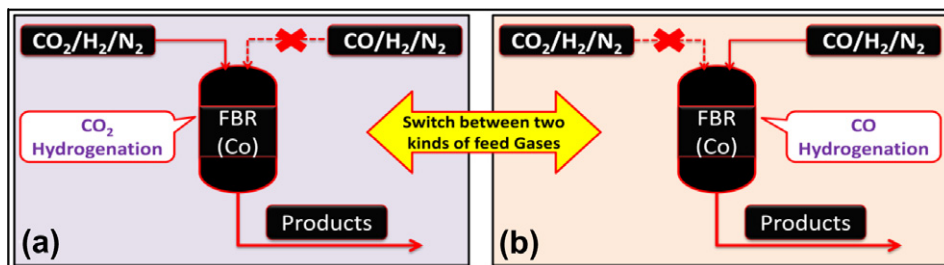


Fig. 1. Simplified flow scheme for FTS using a micro-FBR by switching between $\text{CO}_2/\text{H}_2/\text{N}_2$ and $\text{CO}/\text{H}_2/\text{N}_2$ syngases over a cobalt-based catalyst (reaction conditions as shown in Table 1).

Table 1

Summary of experimental conditions for FTS by switching between CO₂/H₂/N₂ and CO/H₂/N₂ syngases.

Reactor	A micro-FBR
Reactor diameter (mm)	8.0
Catalyst	Co (10 wt.%)/TiO ₂
Catalyst particle size (mm)	0.5–1.0
Catalyst weight (g)	1.0
Catalyst bed height (mm)	27.0
Catalyst bulk density (g/cm ³)	0.88
CO ₂ feed	CO ₂ /H ₂ /N ₂ 23%/67%/10%
CO feed	CO/H ₂ /N ₂ 30%/60%/10%
Total pressure (bar gauge)	20
Flow rate (ml(NTP)/(min-gcat))	30
Temperature (°C)	180–220

- In comparison with CO hydrogenation, CO₂ hydrogenation produces methane rich short chain paraffins, a result in agreement with published research findings [13,15,18,19]. The CH₄

selectivity altered between 87% and 95% when the temperature was increased from 180 °C to 220 °C. The highest CH₄ selectivity was achieved during the initial run at 180 °C. The selectivity decreased when the temperature was increased from 200 °C to 220 °C.

- The C₂₊ selectivity was only around 5%–13%, although it rose in parallel with incremental increases in temperature in the temperature range 200–220 °C.

It is quite interesting to note that the CO₂ conversion achieved its highest value with the reaction temperature at 180 °C (see Fig. 2a) when the CO₂ feed mixture was first introduced into the reactor. After this, when the repeated switching of the feed gas from CO₂ to CO and then back again was initiated, the conversion of both feed gases remained constant. This indicates that the catalyst was not de-activated under those reaction conditions.

It should be noted that when the CO and CO₂ feeds were repeatedly alternated, we could observe no catalyst deactivation at the reaction temperatures of 180 °C and 200 °C. However, we found

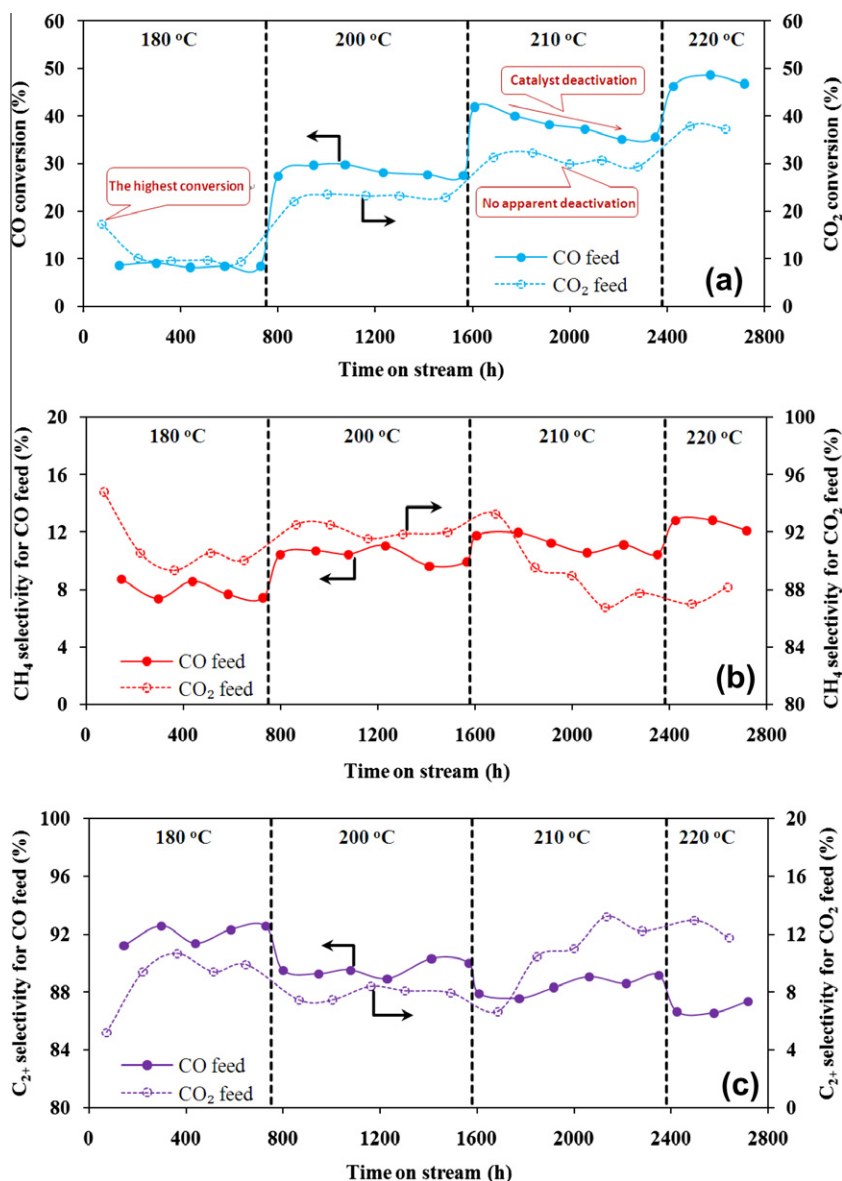


Fig. 2. CO and CO₂ conversion (a); methane selectivity (b); and C₂₊ selectivity (c), as functions of time on stream over a Co/TiO₂ catalyst (reaction conditions as shown in Table 1).

that the CO conversion for the CO feed dropped slightly with time on stream at a reaction temperature of 210 °C, whereas that of the CO₂ feed did not.

3.1.2. Reactant consumption rate and product formation rate

The CO and CO₂ reaction rates for both CO and CO₂ hydrogenation as a function of time on stream are given in Fig. 3a, which show trends similar to those illustrated in Fig. 2a. The CO₂ reaction rate achieved its highest rate when the CO₂ feed was initially introduced into the FBR. This was even higher than the CO reaction rate at 180 °C. However, when the feed gas was subsequently switched from the CO₂ feed to the CO feed and then back again, the CO₂ reaction rate fell to around two times lower than that obtained in the first run. It therefore became necessary to seek more detailed information on what occurred during that period which will be addressed in Section 3.2.

Fig. 3b and c plot the CH₄ and C₂₊ formation rates for both CO and CO₂ feeds. The CH₄ formation rate for CO₂ hydrogenation was far higher than that of CO hydrogenation (Fig. 3b). When the

temperature was increased from 180 °C to 220 °C, the CH₄ formation rate rose for both CO and CO₂ hydrogenation. There was a reduction in the CH₄ rate at 210 °C for both CO and CO₂. The initial run gave the highest CH₄ reaction rate for CO₂ hydrogenation, which was around two times greater than in the other four runs carried out at 180 °C.

It is note that although the C₂₊ formation rate for CO₂ feed improved when the temperature rose, the values of the rate were dramatically lower than those obtained with the CO feed (Fig. 3c). In addition, a drop in the C₂₊ rate with time on stream occurred only in the case of the CO feed when catalyst deactivation took place at 210 °C (Fig. 3a).

3.1.3. Olefin and paraffin formation rates

The light olefin and paraffin formation rates during the switching between the two feed gases as a function of time on stream are plotted in Fig. 4a–c. The first of these, (a), shows the olefin formation rate for the CO feed as fairly constant at each of the reaction temperatures, with a rise in the rate corresponding with each

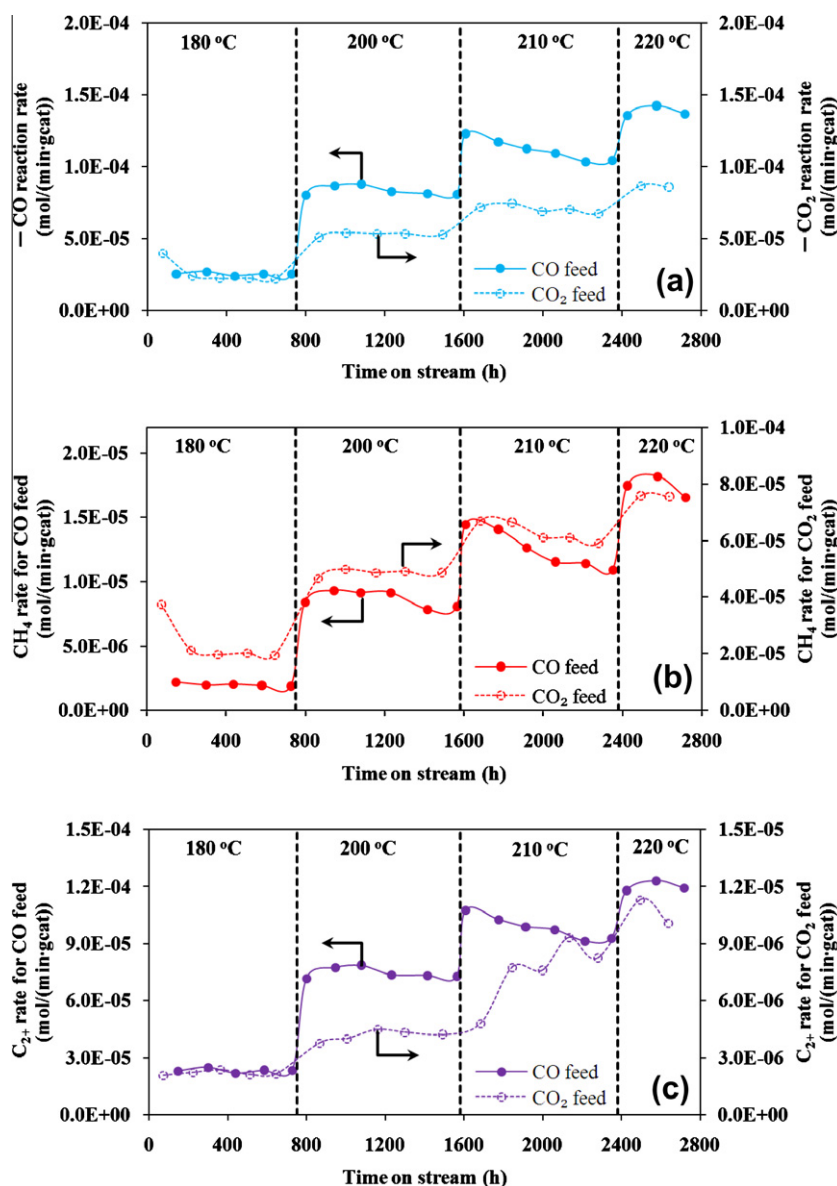


Fig. 3. Reactant consumption rate (a); CH₄ formation rate (b); and C₂₊ formation rate (c), as functions of time on stream over a Co/TiO₂ catalyst (reaction conditions as shown in Table 1).

upward adjustment to the temperature. The paraffin formation rate for CO feed also rose when the temperature increased, but showed a visible decline with the accumulation of time on stream at 210 °C (Fig. 4b). Comparing the results we obtained for CO hydrogenation with those for CO₂ hydrogenation, we found that no olefin could be detected in the latter, and that all the products

were paraffins. The paraffin formation rate for the CO₂ feed was remarkably constant at each of the reaction temperatures, even at the higher temperatures (210 °C and 220 °C), which differed from the case of CO hydrogenation (Fig. 4c).

In summary, when the catalyst was deactivated at 210 °C, for the CO feed only the paraffin product formation rate

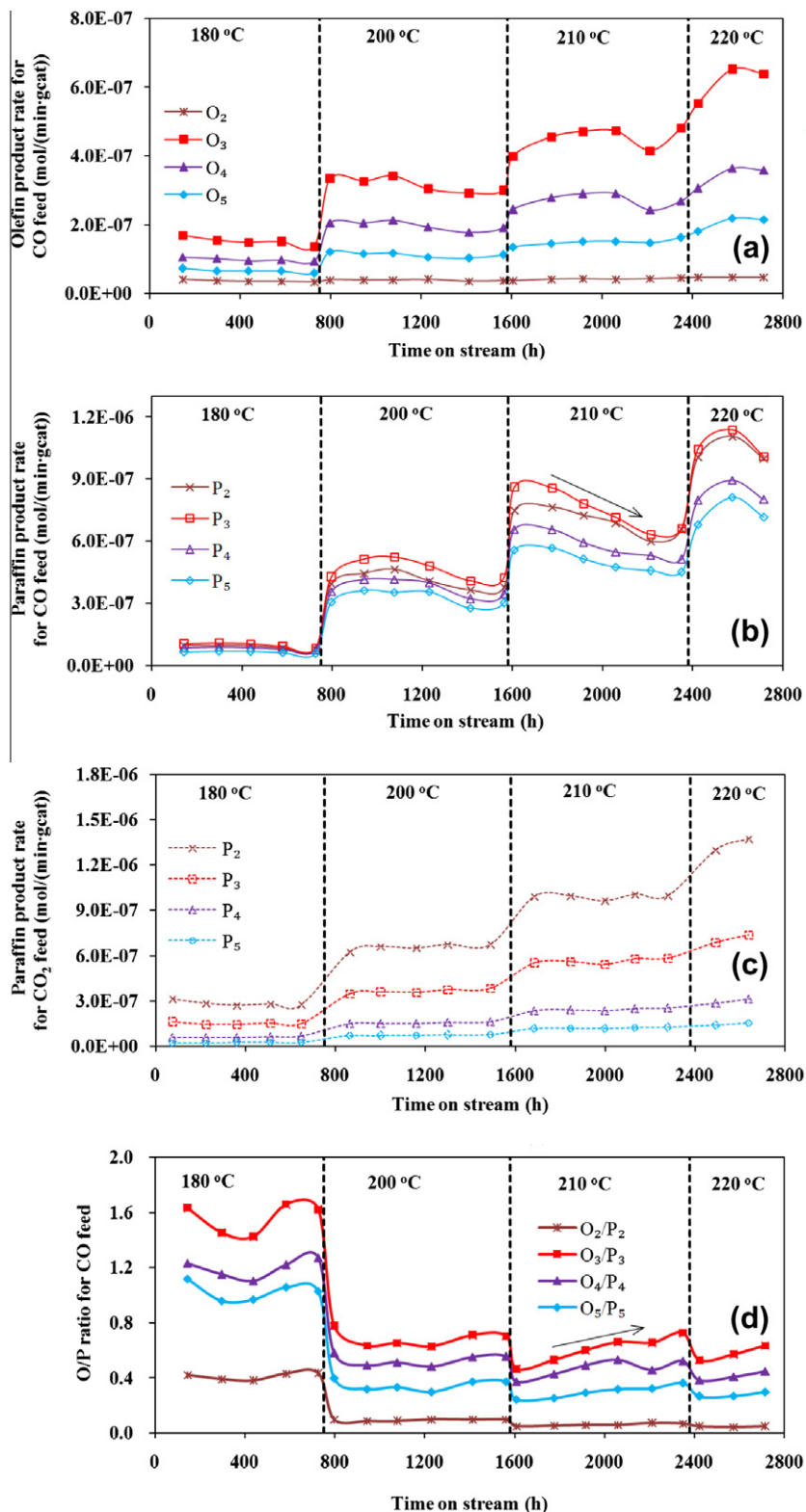


Fig. 4. The olefin formation rate for CO feed (a); paraffin formation rate for CO feed (b); paraffin formation rate for CO₂ feed (c); and olefin/paraffin (O/P) ratio for CO feed (d), as functions of time on stream for a Co/TiO₂ catalyst (reaction conditions as shown in Table 1).

showed a significant decline, while the olefin rate did not; but for the CO₂ feed, with the exception of CH₄, the paraffin product formation rate was not affected at that reaction temperature.

In addition, the typical pattern of relatively low yields of ethene and ethane was obtained for the CO feed [26–29], but in contrast the amount of ethane produced by the CO₂ feed was greater than for the other hydrocarbons with a chain length of $n > 2$.

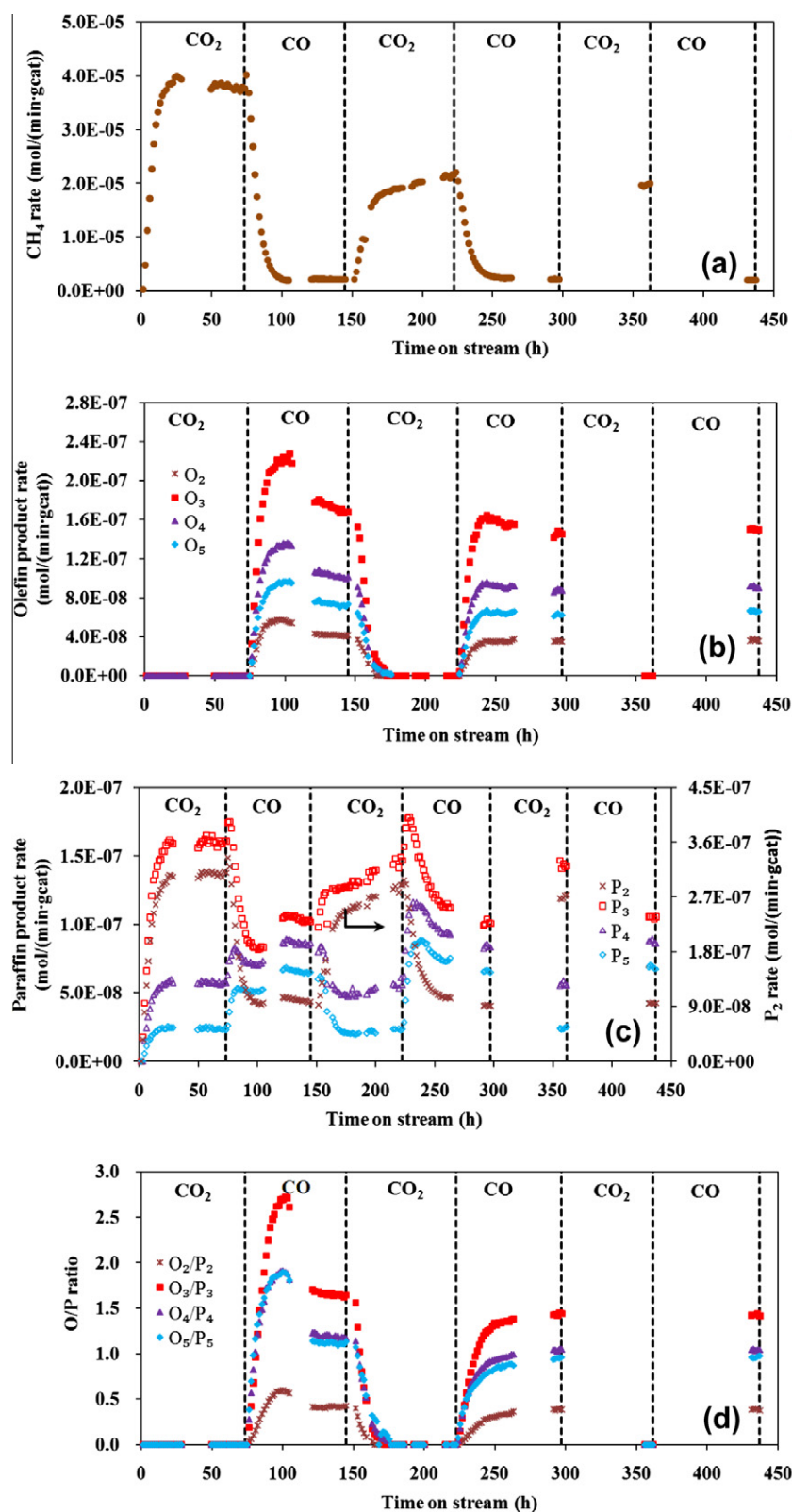


Fig. 5. The CH₄ formation rate (a); olefin formation rate (b); paraffin formation rate (c); and olefin/paraffin (O/P) ratio for CO feed (d), as functions of time on stream for a Co/TiO₂ catalyst at 180 °C, 20 bar (gauge) and 30 ml(NTP)/(min·gcat): CO₂ represents the CO₂ feed (CO₂/H₂/N₂ = 23%/67%/10%) and CO refers to the CO feed (CO₂/H₂/N₂ = 30%/60%/10%).

Because all the products of the CO₂ feed were saturated paraffins, only the olefin to paraffin (O/P) ratio derived from the CO feed is given in Fig. 4d. It is generally accepted [30,31] that the O/P ratio changes as a function of carbon number, and that an increment in the carbon number causes a drop in the O/P ratio under each reaction condition excluding O₂/P₂. It should be noted that the O/P ratio did not change much with time on stream at the lower temperature of 180 °C, as well as at 200 °C. However, an increase in the O/P ratio was obtained with time on stream at 210 °C and 220 °C, respectively.

3.2. The effect of CO₂ on the catalyst activity and selectivity for CO and CO₂ hydrogenations at 180 °C

Fig. 2a shows that: CO conversion was quite stable when switching occurred between CO and CO₂ feeds in the FBR at a constant temperature of 180 °C; however, the CO₂ conversion was initially changed by the introduction of the CO feed into the reactor, but then became stable. Detailed information concerning the product formation rate and O/P ratio is given in Fig. 5. When we compared the CO₂ hydrogenation data recorded both before and after the initial CO FTS at 180 °C, we found that:

- the CH₄ formation rate decreased from 3.8E-05 mol/(min·gcat) in the first run of CO₂ hydrogenation to 2.1E-05 mol/(min·gcat) in the second (around two times lower than the first);
- the P₂ and P₃ rates decreased slightly, from 3.1E-07 mol/(min·gcat) to 2.8 E-07 mol/(min·gcat) for P₂ and from 1.6E-07 mol/(min·gcat) to 1.5E-07 mol/(min·gcat) for P₃;
- the P₄ and P₅ rates remained similar to the values obtained in the first run of the CO₂ feed.

With the subsequent repetition of alternation between the two feeds, we found that each of the paraffin formation rates maintained a pseudo-stable state.

When the CO feed was initially introduced into the reactor, the time on stream was 73 h. Between that point and 144 h:

- each of the olefin products reached its highest rate and then decreased to a stable state, as shown in Fig. 5b;
- each of the paraffin products achieved its lowest rate, and then increased to a stable state (see Fig. 5c); and
- each of the O_n/P_n ratios attained its maximum value, and then dropped to a stable state value (see Fig. 5d).

The data revealed that both the olefin and paraffin formation rates were suddenly changed from one pseudo-stable state to another during the initial run of CO hydrogenation. With subsequent repeated switching between the two feeds, we found that both the paraffin and olefin product formation rates and O/P ratio for the CO feed reverted to the values obtained with the time on stream from 120 to 144 h as shown in Fig. 5.

4. Discussion

Because that the catalyst was not apparently deactivated under reaction conditions at 180 °C for both CO and CO₂ hydrogenations, deactivation cannot explain the observed phenomenon that the catalyst activity and product selectivity for CO₂ hydrogenation was initially changed by introducing the CO feed into the reactor at the reaction temperature of 180 °C (Figs. 2, 3 and 5).

One of the possible reasons for the observed significant changes during the first switching is the formation of the liquid phase either on the catalyst surface or in the catalyst pores. Some

researchers [32–35] have reported that under typical reaction conditions the FT products distribute between the vapor and liquid phases within the reactor. Lu et al. [36,37] concluded that the liquid product deposition in the catalyst can change the catalyst activity and product selectivity. Furthermore, by using the deuterium tracer, Liu and co-workers [38] measured that the product accumulation in FTS occurs not only in large continuously stirred tank reactors, but also in small FBR.

Grounding our reasoning on the research reported in the literature and our own experimental results, we postulate that liquid products may be deposited on the catalyst surface or in the catalyst pores during CO FTS. For our experiments, when the CO₂ feed was initially introduced into the reactor, the dominant products were methane-rich short chain paraffins, so that only a gas phase occurred. Then, when we switched the feed gas to a CO syngas, which would gradually replace the CO₂ in the reactor, long chain waxes formed, and accumulated to form a liquid phase on the catalyst surface and in the catalyst pores. After that, when we switched back to the CO₂ feed, it is possible that a certain amount of liquid remained on the catalyst surface or in the catalyst pores during CO₂ hydrogenation. These liquid products could therefore change the mass transfer of reactants and products, and, further, affect the catalyst activity and product selectivity.

FTS is a surface catalytic reaction. The H/C (hydrogen to carbon) concentration on the catalyst surface can affect the product selectivity of both of the CO and CO₂ hydrogenations [13,39]. A low H/C ratio leads to reduced selectivity of short chain hydrocarbons, high selectivity of long chain hydrocarbons and olefins, and a low selectivity of paraffins. The absorption of CO, CO₂ and H₂ on the catalyst surface is likely to be different for a gas or a liquid phase in the catalyst pores, which will also affect the H/C surface ratio. As shown in Figs. 2 and 5, the results from the first run of the CO₂ hydrogenation with those of the second, we observe that in the latter the CO₂ conversion and methane selectivity dropped and C₂₊ selectivity rose. These data suggest that a lower H/C ratio is obtained on a liquid-covered catalyst surface than on the dry catalyst surface. This means that the liquid phase is more favorable to the absorption of CO₂ than H₂ for CO₂ hydrogenation.

Because that a lower O/P ratio was obtained when the olefin and paraffin formation rates were suddenly changed from one pseudo-state to another during initially introducing the CO feed into the reactor as shown in Fig. 5, a higher H/C ratio is obtained on a liquid-covered catalyst surface than on a dry catalyst surface for CO hydrogenation. This finding is in agreement with the claim made by Lu et al. [37] that the H₂/CO ratio in the liquid-covered catalyst is far higher than in the feed gas.

Fig. 6 shows the olefin, paraffin and combined olefin + paraffin formation rates with carbon numbers equal to 3 (a) and 6 (b) with time on stream for the first 145 h. As can be seen, with time on stream the olefin formation rate (O₃ and O₆) decreased between the first pseudo-state and the second, while at the same time the paraffin formation rate (P₃ and P₆) increased, the total (P₃ + O₃) rate dropped slightly, and the (P₆ + O₆) rate remained constant. These data indicate that the liquid phase may promote a secondary reaction of the primary olefins, especially in the case of olefin hydrogenation (see Reaction (a) below).



Because the H/C ratio in the liquid phase is higher than in the gas phase for CO feed, olefin hydrogenation will more likely occur in the liquid phase with the re-adsorbed olefin products.

When mild catalyst deactivation was observed at a time on stream of 1600–2700 h and reaction temperatures of 210 °C and 220 °C (Figs. 2 and 3), a small amount of CO₂ product was detected in the tailgas for CO hydrogenation (see Fig. 7). There are two ways

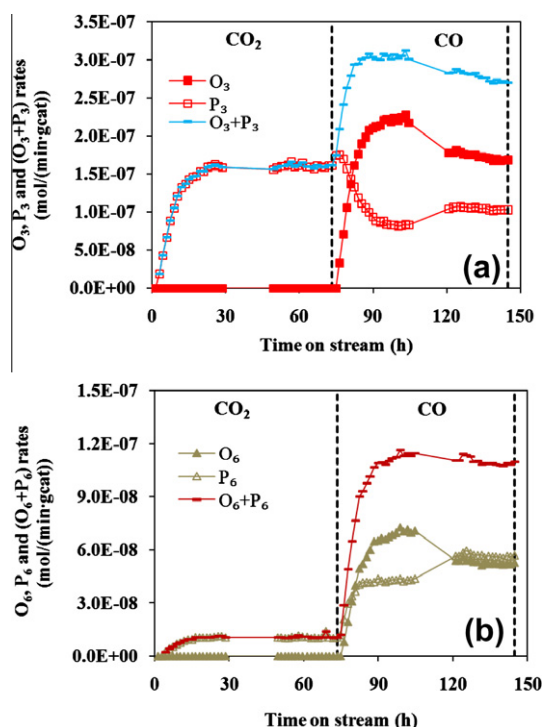
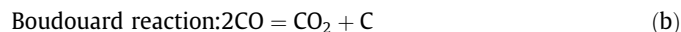


Fig. 6. Olefin, paraffin and combined olefin + paraffin formation rates for carbon numbers equal to 3 (a) and 6 (b) as functions of time on stream for a Co/TiO₂ catalyst at 180 °C, 20 bar (gauge) and 30 ml(NTP)/(min·gcat). CO₂ represents the CO₂ feed (CO₂/H₂/N₂ = 23%/67%/10%) and CO refers to the CO feed (CO₂/H₂/N₂ = 30%/60%/10%).

to produce CO₂ during CO hydrogenation. These can be expressed in terms of the reaction (b) and (c):

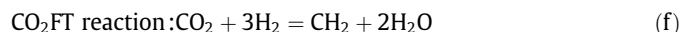
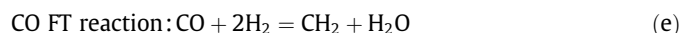
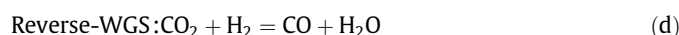


The Boudouard reaction provides atomic carbon, which may be transformed into more stable species (such as bulk cobalt carbide and polymeric carbon), and negatively influences FTS activity [6,35]. However, Dry [40] reported that little or no carbon was deposited on the catalyst surface at an FT operating temperature below about 240 °C, regardless of whether Ni, Co, Ru or Fe-based catalysts were being used. The experiments we conducted were in the range 180–220 °C, which is lower than 240 °C. Alternatively, if the catalyst deactivation we observed in the experiments was caused by carbon deposition, the catalyst activity for both CO and CO₂ hydrogenations should have decreased. However, the data in

both Figs. 2 and 3a shows that there was no deactivation evident for the CO₂ feed with the time on stream at an operating temperature of 210 °C. Thus, we suggest that carbon deposits were unlikely to be the main reason for the deactivation of the catalyst under the reaction conditions we chose.

Therefore, we have to consider whether the production of CO₂ is caused mainly by WGS rather than the Boudouard reaction. It is widely accepted that the WGS reaction is attributable to the transformation of cobalt to oxide forms [6], in other words to re-oxidation of the catalyst, which is the one of the reasons for cobalt catalyst deactivation [6,8]. When the operating temperature was increased from 200 °C to 210 °C, the feed gas was the CO based syngas. The CO₂ product was the first to be detected (Fig. 7), which indicates that the catalyst re-oxidation was triggered by the CO feed. Dalai and Davis [8] suggested that the effect of water on a supported cobalt catalyst can be viewed as starting an oxidation process, and that the extent of oxidation is a function of cobalt crystallite size and the ratio of the partial pressures of water and hydrogen (P_{H₂O}/P_{H₂}) in the reactor. The water concentration at the outlet of the reactor for CO feed was also plotted in Fig. 7, which shows that the water concentration was increased with an increase of the temperature and the CO₂ selectivity curve was similar to that for the H₂O concentration at the outlet at 210 °C and 220 °C. There was no evidence to show that the catalyst re-oxidation could be attributed to the CO₂ feed in our experimental system.

Based on the discussion above, we suggest that the main cause of the mild catalyst deactivation we observed was catalyst re-oxidation by water. The cobalt oxide forms have activity for both the WGS and reverse WGS reactions (d).



We postulate that once the CO₂ feed was switched back into the reactor again after the cobalt catalyst re-oxidized, the reverse-WGS reaction (d) could develop, and produce CO. The CO produced would subsequently react in the CO FT reaction (e) to form organic products, which could in turn increase the conversion of CO₂ to CO. Although the CO₂ FT reaction (f) rate will be affected by catalyst re-oxidation, the reverse-WGS reaction explains why the total CO₂ conversion for the CO₂ feed was not markedly reduced when the catalyst was deactivated with the time on stream (Fig. 2a).

As previously mentioned, the liquid phase on the catalyst surface and in the catalyst pores may affect the catalyst activity and product selectivity. For instance, secondary olefin hydrogenation can be fostered by the liquid phase (Fig. 6). Any of the factors that

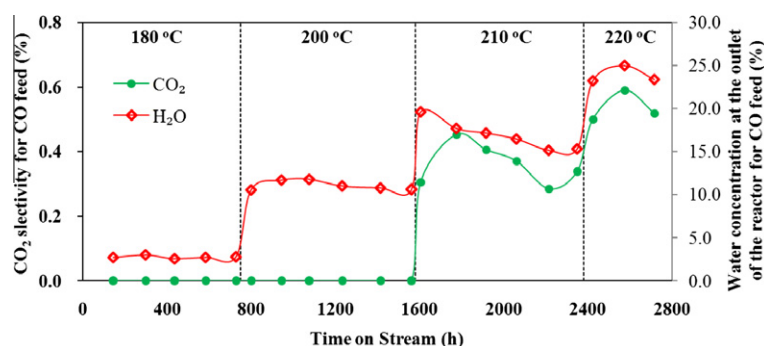


Fig. 7. CO₂ selectivity and water concentration at the outlet of the reactor for CO hydrogenation as a function of time on stream for a Co/TiO₂ catalyst (reaction conditions as shown in Table 1).

alter the amount of liquid accumulated on the catalyst surface or in the catalyst pores will affect the FT product composition as well. The catalyst deactivation observed in our experiments reduced the CO conversion, and could then decrease the amount of the liquid that accumulated in the reactor. Therefore, the extent of the secondary reaction could be restricted by catalyst deactivation, which means the rate of reaction (a) (olefin hydrogenation) would be decreased with a reduced CO conversion during FTS. Thus, it is found that the olefin product formation rate did not decrease, but the paraffin product formation rate did with an increase of the O/P ratio with time on stream at the temperature of 210 °C and 220 °C (Fig. 4), respectively. These phenomenons also support the hypothesis of the presence of a liquid phase in the reactor.

5. Conclusions

For the experimental work described in this paper, we prepared 10 wt.% Co/TiO₂ catalyst by impregnating TiO₂ with a cobalt nitrate solution. The new series of experiments were conducted by switching repeatedly between a CO feed (CO/H₂/N₂ = 30%/60%/10%) and a CO₂ feed (CO₂/H₂/N₂ = 23%/67%/10%) in a FBR at 180–220 °C, 20 bar (gauge) and 30 ml(NTP)/(min·gcat).

We have shown that both CO and CO₂ are readily hydrogenated over a cobalt-based catalyst, and that, unlike CO hydrogenation, CO₂ hydrogenation produced methane-rich short chain saturated hydrocarbons. When we continually alternated the CO and CO₂ feeds, we were unable to find any signs of catalyst deactivation at reaction temperatures of 180 °C and 200 °C.

The data show that dramatic changes occurred in the catalyst activity and product selectivity for CO₂ hydrogenation between the initial CO FTS at 180 °C and the succeeding run. In addition, during the initial CO hydrogenation on the cobalt catalyst, both the olefin and paraffin formation rates suddenly changed from one pseudo-stable state to another, with a higher O/P ratio obtained initially. According to our own data and the findings published in the relevant scientific literature review, we concluded that these changes could be attributed to liquid products deposited on the catalyst surface or in the catalyst pores during CO FTS.

A mild catalyst deactivation was observed at the operating temperatures of 210 °C and 220 °C, respectively. During the period when the catalyst was deactivated, we found that: (1) the reaction rate decreased for CO hydrogenation rather than for CO₂ hydrogenation; (2) only the paraffin product formation rate dropped significantly, while there was no decline in the yield of olefin for CO hydrogenation; (3) except for CH₄, the paraffin product formation rate for the CO₂ feed was not reduced; (4) and CO₂ product was detected for CO hydrogenation. These suggest that the catalyst deactivation was caused by the re-oxidation of the cobalt catalyst by water.

Although we could not explain the experimental phenomena fully, we must emphasize that the long term effect of CO₂ on the properties of a cobalt-based catalyst was very small under the given reaction conditions. We also confirmed that it may not be necessary to remove CO₂ from the raw syngas for cobalt-based FTS. These results provide some guidance on how to design FTS processes and FT catalysts to improve product selectivity.

Acknowledgements

The authors are grateful for the support received from Golden Nest International (Pty) Ltd., the National Research Foundation (NRF), the South African Research Chairs Initiative (SARChI), the Technology and Human Resources for Industry Programme (THRIP), the University of the Witwatersrand, Johannesburg, and the South African National Energy Research Institute (SANERI).

References

- [1] M.E. Dry, The Fischer–Tropsch process: 1950–2000, *Catal. Today* 71 (2002) 227–241.
- [2] H. Schulz, Short history and present trends of Fischer–Tropsch synthesis, *Appl. Catal. A: Gen.* 186 (1999) 3–12.
- [3] O.O. James, A.M. Mesubi, T.C. Ako, S. Maity, Increasing carbon utilization in Fischer–Tropsch synthesis using H₂-deficient or CO₂-rich syngas feeds, *Fuel Process. Technol.* 91 (2010) 136–144.
- [4] G. Arzamendi, P.M. Dieguez, M. Montes, J.A. Odriozola, E.F. Sousa-Aguiar, L.M. Gandia, Computational fluid dynamics study of heat transfer in a microchannel reactor for low-temperature Fischer–Tropsch synthesis, *Chem. Eng. J.* 160 (2010) 915–922.
- [5] R.L. Espinoza, A.P. Steynberg, B. Jager, A.C.V. Osloo, Low temperature Fischer–Tropsch synthesis from a Sasol perspective, *Appl. Catal. A: Gen.* 186 (1999) 13–26.
- [6] N.E. Tsakoumis, M. Ronning, O. Borg, E. Rytter, A. Holmen, Deactivation of cobalt based Fischer–Tropsch catalysts: a review, *Catal. Today* 154 (2010) 162–182.
- [7] J. van de Loosdrecht, B. Balzhinimaev, J.A. Dalmon, J.W. Niemantsverdriet, S.V. Tsybulya, A.M. Saib, P.J. van Berge, J.L. Visagie, Cobalt Fischer–Tropsch synthesis: deactivation by oxidation?, *Catal Today* 123 (2007) 293–302.
- [8] A.K. Dalai, B.H. Davis, Fischer–Tropsch synthesis: a review of water effects on the performances of unsupported and supported Co catalysts, *Appl. Catal. A: Gen.* 348 (2008) 1–15.
- [9] E. van Steen, M. Claeys, M.E. Dry, J. van de Loosdrecht, E.L. Viljoen, J.L. Visagie, Stability of nanocrystals: thermodynamic analysis of oxidation and re-reduction of cobalt in water/hydrogen mixtures, *J. Phys. Chem. B* 109 (2005) 3575–3577.
- [10] E. Iglesia, Design, synthesis, and use of cobalt-based Fischer–Tropsch synthesis catalysts, *Appl. Catal. A: Gen.* 161 (1997) 59–78.
- [11] J. Li, G. Jacobs, T. Das, B.H. Davis, Fischer–Tropsch synthesis: effect of water on the catalytic properties of a ruthenium promoted Co/TiO₂ catalyst, *Appl. Catal. A: Gen.* 233 (2002) 255–262.
- [12] C. Higman, M. Burgt, *Gasification*, Guif Professional Publishing, Burlington, 2003.
- [13] C.G. Visconti, L. Lietti, E. Tronconi, P. Forzatti, R. Zennaro, E. Finocchio, Fischer–Tropsch synthesis on a Co/Al₂O₃ catalyst with CO₂ containing syngas, *Appl. Catal. A: Gen.* 355 (2009) 61–68.
- [14] M.K. Gnanamani, W.D. Shafer, D.E. Sparks, B.H. Davis, Fischer–Tropsch synthesis: Effect of CO₂ containing syngas over Pt promoted Co/γ-Al₂O₃ and K-promoted Fe catalysts, *Catal. Commun.* 12 (2011) 936–939.
- [15] Y. Zhang, G. Jacobs, D.E. Sparks, M.E. Dry, B.H. Davis, CO and CO₂ hydrogenation study on supported cobalt Fischer–Tropsch synthesis catalysts, *Catal. Today* 71 (2002) 411–418.
- [16] S.M. Kim, J.W. Bae, Y.J. Lee, K.W. Jun, Effect of CO₂ in the feed stream on the deactivation of Co/γ-Al₂O₃ Fischer–Tropsch catalyst, *Catal. Commun.* 9 (2008) 2269–2273.
- [17] T. Riedel, G. Schaub, Low-temperature Fischer–Tropsch synthesis on cobalt catalysts-effects of CO₂, *Top. Catal.* 26 (2003) 145–156.
- [18] T. Riedel, M. Claeys, H. Schulz, G. Schaub, S.S. Nam, K.W. Jun, M.J. Choi, G. Kishan, K.W. Lee, Comparative study of Fischer–Tropsch synthesis with H₂/CO and H₂/CO₂ syngas using Fe- and Co-based catalysts, *Appl. Catal. A: Gen.* 186 (1999) 201–213.
- [19] Y. Yao, D. Hildebrandt, D. Glasser, X. Liu, Fischer–Tropsch synthesis using H₂/CO₂ syngas mixtures over a cobalt catalyst, *Ind. Eng. Chem. Res.* 49 (2010) 11061–11066.
- [20] W.W. Russell, G.H. Miller, Catalytic hydrogenation of carbon dioxide to higher hydrocarbons, *J. Am. Ceram. Soc.* 72 (1949) 2446–2454.
- [21] A.N. Akin, M. Ataman, Z.E. Aksoylu, Z.I. Onsana, CO₂ Fixation by hydrogenation over co-precipitated Co/Al₂O₃, *React. Kinet. Catal. Lett.* 76 (2002) 265–270.
- [22] R. Bai, Y. Tan, Y. Han, Study on the carbon dioxide hydrogenation to iso-alkanes over Fe–Zn–M/zeolite composite catalysts, *Fuel Process. Technol.* 86 (2004) 293–301.
- [23] T. Herranz, S. Rojas, F.J. Perez-Alonso, M. Ojeda, P. Terreros, J.L.G. Fierro, Hydrogenation of carbon oxides over promoted Fe–Mn catalysts prepared by the microemulsion methodology, *Appl. Catal. A: Gen.* 311 (2006) 66–75.
- [24] M.E. Dry, The Fischer–Tropsch synthesis, in: J.R. Anderson, M.E. Boudart (Eds.), *Catalysis Science and Technology*, Springer, New York, 1981, pp. 159–255.
- [25] J. Li, The preparation, characterization and evaluation of boron-modified Co/TiO₂ Fischer–Tropsch catalysts, PhD thesis, University of the Witwatersrand, Johannesburg, South Africa, 1999.
- [26] B.W. Wojciechowski, The kinetics of the Fischer–Tropsch synthesis, catalysis reviews, *Sci. Eng.* 30 (1988) 629–702.
- [27] T. Komaya, A.T. Bell, Estimates of rate coefficients for elementary processes occurring during Fischer–Tropsch synthesis over Ru/TiO₂, *J. Catal.* 146 (1994) 237–248.
- [28] J. Cheng, P. Hu, P. Ellis, S. French, G. Kelly, C.M. Lok, A DFT study of the chain growth probability in Fischer–Tropsch synthesis, *J. Catal.* 257 (2008) 221–228.
- [29] S. Novak, R.J. Madon, H. Suhl, Models of hydrocarbon product distributions in Fischer–Tropsch synthesis, *J. Chem. Phys.* 74 (1981) 6083–6091.
- [30] E.W. Kuipers, I.H. Vinkenburg, H. Oosterbeek, Chain length dependence of α-olefin re-adsorption in Fischer–Tropsch synthesis, *J. Catal.* 152 (1995) 137–146.
- [31] B. Shi, B.H. Davis, Fischer–Tropsch synthesis: the paraffin to olefin ratio as a function of carbon number, *Catal. Today* 106 (2005) 129–131.

- [32] J.J. Marano, G.D. Holder, Characterization of Fischer–Tropsch liquids for vapor–liquid equilibria calculations, *Fluid Phase Equilib.* 138 (2007) 1–21.
- [33] L. Caldwell, D.S. van Vuuren, On the formation and composition of the liquid phase in Fischer–Tropsch reactors, *Chem. Eng. Sci.* 41 (1986) 89–96.
- [34] C.N. Satterfield, H.G. Stenger Jr., Effect of liquid composition on the slurry Fischer–Tropsch synthesis: 1. Rate of reaction, *Ind. Eng. Chem. Process Des. Dev.* 24 (1985) 407–411.
- [35] C.N. Satterfield, H.G. Stenger Jr., Effect of liquid composition on the slurry Fischer–Tropsch synthesis: 2. Product selectivity, *Ind. Eng. Chem. Process Des. Dev.* 24 (1985) 411–415.
- [36] X. Lu, D. Hildebrandt, X. Liu, D. Glasser, Making sense of the Fischer–Tropsch synthesis reaction: start-up, *Ind. Eng. Chem. Res.* 49 (2010) 9753–9758.
- [37] X. Lu, X. Zhu, D. Hildebrandt, X. Liu, D. Glasser, New way to look at Fischer–Tropsch synthesis using flushing, *Ind. Eng. Chem. Res.* 50 (2011) 4359–4365.
- [38] Y. Liu, S. Zheng, B. Shi, J. Li, Deuterium tracer study of Fischer–Tropsch synthesis: a method to eliminate accumulation problems, *J. Mol. Catal. A: Chem.* 276 (2007) 110–115.
- [39] E. Zagli, J.L. Falconer, Carbon dioxide adsorption and methanation on ruthenium, *J. Catal.* 69 (1981) 1–8.
- [40] M.E. Dry, FT catalysts, in: A. Steynberg, M.E. Dry (Eds.), *Studies in Surface Science and Catalysis*, vol. 152, Elsevier, New York, 2004, pp. 533–600.